Engineering Applications of Artificial Intelligence 158 (2025) 111315

Contents lists available at ScienceDirect

Engineering Applications of Artificial Intelligence

journal homepage: www.elsevier.com/locate/engappai

Research paper

Lightweight binary convolutional-transformers fusion network for facial expression recognition

Xing Jin [a,*], Xiyin Wu [b], Libo Weng [c], Qiaolin Ye [a]

[a] College of Information Science and Technology & Artificial Intelligence, Nanjing Forestry University, Nanjing, 210037, Jiangsu, China
[b] College of Big Data and Information Engineering, Guizhou University, Guiyang, 550025, Guizhou, China
[c] College of Computer Science and Technology, Zhejiang University of Technology, Hangzhou, 310023, Zhejiang, China

ARTICLE INFO

Keywords:
Facial expression recognition
Lightweight network
Binary neural network
Vision transformer

ABSTRACT

Deep learning-based methods exploit local facial regions for facial expression recognition (FER) but overlook long-range dependencies of facial muscle movements. Moreover, deploying existing large-scale deep networks on mobile devices remains a huge challenge. To this end, this paper designs a mini-binary transformer (MiniBTR) for FER. Specifically, we first construct image patches by regions of interest (ROIs) which are associated with facial action units (AUs) and represent these patches by the histogram of oriented gradient (HOG) feature. Then, by integrating the complementary advantages of convolutional layers (Convs) and multi-head self-attention mechanism (MSA), an effective and efficient mini-transformer (MiniTR) architecture with 69K parameters is proposed to follow the human–machine collaborative strategy. Finally, we design the binary model MiniBTR by extending MiniTR with a binary operation. Extensive experimental results on four publicly available datasets demonstrate that the MiniBTR yields comparable results in terms of recognition accuracy, model size and inference speed while offering a more deployable yet high-performing alternative.

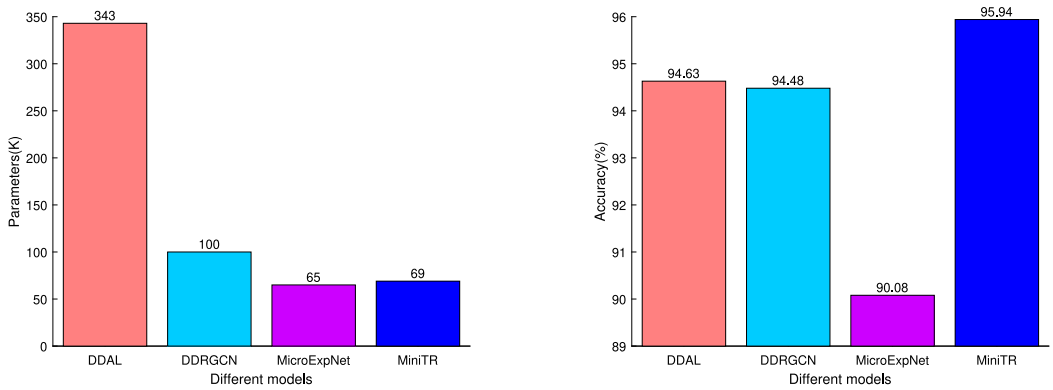
1. Introduction

Facial expression recognition (FER) has achieved remarkable success in many applications, such as consumer behavior analysis, online education and health-care (Lu et al., 2025; Tutuianu et al., 2024). To construct an efficient emotional analysis system, some earlier works extract features by linear discriminant analysis (Zhu et al., 2022; Ye et al., 2016) or dictionary learning, and then combine different classifiers, i.e., support vector machine, Bayes classifier. These models suffer a lower recognition rate. To alleviate this issue, deep learning-based facial expression recognition methods were proposed and achieved remarkable results (Xiong et al., 2024; Zhu et al., 2025). To this end, Zhang et al. (2021) proposed a real-time emotion recognition system. However, this model suffers more calculation complexity by exploiting reinforcement learning and domain knowledge. Moreover, Li et al. (2021) designed an informative and discriminative feature learning framework for FER with 89 MB parameters. So the mobile deployment of facial expression recognition models has attracted great interest over the past decade (Lan et al., 2025; Pan et al., 2025), and deploying existing deep models on resource-limited mobile platforms remains a huge challenge.

Many researchers designed lightweight FER systems mainly from two directions (Zhang et al., 2025). One direction is to reduce the depth of the deep network. Lee et al. (2020) proposed the EmotionNet Nano network which only has nineteen layers and leverage residual architecture for FER. More recently, some lightweight deep networks, i.e., DDAL (Jin et al., 2020), DDRGCN (Jin et al., 2021), MicroExpNet (Cugu et al., 2019), designed for FER with fewer layers. The comparison results of the proposed MiniTR and other deep models are shown in Fig. 1. Our proposed MiniTR obtains the highest accuracy with 69K parameters. Another direction is to reduce network input. Shan et al. (2005) used the 14×19 facial region as input. Although the whole facial image has become the standard input, the preprocessing procedure is more complex and the background noise is easily introduced. Based on relevant research on facial action coding system (FACS) (Ekman, 1997; Mao et al., 2025), different facial expressions are generated by combinations of facial local muscle movements defined as action units (AUs). AUs-based methods (Danelakis et al., 2018; Jin et al., 2021), which try to exploit some specific facial local regions for FER, have become research hotspots. Using certain facial local regions can reduce background noise and the computational cost of the input. In addition, these AUs are related to the muscles in different positions of the face. However, the classic convolutional layers (Convs) operation cannot effectively model long-range dependencies between AUs in different facial expressions. Recently, the

* Corresponding author.
E-mail addresses: xingjin@njfu.edu.cn (X. Jin), qqwu3@gzu.edu.cn (X. Wu), wenglibo@zjut.edu.cn (L. Weng), yqlcom@njfu.edu.cn (Q. Ye).

https://doi.org/10.1016/j.engappai.2025.111315
Received 10 July 2024; Received in revised form 14 April 2025; Accepted 26 May 2025
Available online 13 June 2025
0952-1976/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.



(a) NetScore of different models (b) Accuracies of different models

Fig. 1. The parameters and accuracies of different methods on RaFD dataset. (a) shows parameters of different models. Our proposed model mini-transformer (MiniTR) has only 69K parameters. (b) shows the accuracies of different models. While the parameters of the network are gradually reduced from 343K (DDAL Jin et al., 2020) to 65K (MicroExpNet Cugu et al., 2019), the accuracies are also decreased from 94.63% to 90.08%. Compared to other lightweight models, our proposed MiniTR achieves the best accuracy with fewer parameters.

transformers (Brown et al., 2020; Zhou et al., 2025) obtained success in natural language processing and computer vision by exploiting the self-attention mechanism which plays an essential impact in learning long-range dependencies (Bougourzi et al., 2020; Srinivas et al., 2021). Xie et al. (2023) designed a novel Transformer block enhancement module (TBEM) to obtain robust feature representation for facial expression analysis. Jiang et al. (2024) designed a novel lightweight deep model by combining classical MobileNetV2 and transformer to maximize the retention of global and local features. The MobileViT has 972K model parameters and obtains 92.30% accuracy on the RaFD dataset. So combining the Convs and transformer architecture is becoming a hot topic for FER.

Although some lightweight models have obtained impressive results for expression analysis, designing an efficient deep model needs to compromise recognition accuracy, model size and inference speed. Moreover, to deploy deep models on mobile devices, several hardware properties, i.e., memory saving, power efficiency and significant acceleration, need to be considered effectively. Neural network quantization is an effective way to achieve these balances. Recently, binary neural networks (BNNs) enjoyed better hardware-friendly properties and gained substantial interest (Qin et al., 2020). The weights and activations were represented by binary data, which could only have two possible values, i.e., $-1(0)$ or $+1$. Moreover, the traditional matrix multiplication operations are replaced with efficient XNOR-Bitcount operations. The XNOR-Net (Rastegari et al., 2016) can obtain 32× memory saving and 58× speedup on CPUs. Although binary neural networks have achieved promising performance in different tasks, i.e., image classification and object detection, designing an effective binary neural network for FER has attracted increasing attention.

To alleviate the above issues, we propose a novel and lightweight deep model for FER from two perspectives. One is using transformer architecture. The transformer architecture can model long-range muscle movements which play an important role in expression recognition. The other one is lightweight and binarization. The lightweight BNN can improve the time and space efficiency to deploy the model on edge devices. Specifically, we crop facial regions of interest (ROIs) associated with facial AUs. All these ROIs are utilized to construct image patches for the self-attention mechanism. Then, we design a lightweight deep model named Mini-baseline by exploiting the human–machine collaborative network design algorithm (Lee et al., 2020). Moreover, we design the MiniTR architecture by exploiting the complementarity of Convs and the self-attention mechanism, in which a transformer encoder is added at the end of the last Convs of the Mini-baseline. At last, we extend the MiniTR to a binary model. We conducted comparative experiments on four facial expression datasets to validate the effectiveness of the proposed deep networks. Our main contributions are summarized as follows:

(1) We crop facial local regions to construct image patches rather than evenly divide the whole face image for the vision transformer. These local regions are associated with facial AUs and represented by the histogram of oriented gradient (HOG) feature. The advantage of HOG features is to maintain high feature representation ability with fewer dimensions. Moreover, using facial expressions associated with local regions can reduce network input.

(2) We propose a novel mini-transformer (named MiniTR) architecture for FER, in which the transformer encoder is added at the last Convs of the Mini-baseline. Compared to other lightweight deep models, the MiniTR only has 69K model parameters. However, the proposed MiniTR achieves higher performances on three performance indicators, such as recognition accuracy, model size and inference speed.

(3) Benefiting from the characteristics of binarizing, we propose a mini-binary transformer (named MiniBTR) by extending the MiniTR with binarizing. Compared to other lightweight binarized deep networks, the proposed MiniBTR still achieves better recognition accuracy.

2. Related work

In this section, we investigate the use of facial local information and lightweight facial expression recognition networks. Since the vision transformer and binary deep networks play crucial roles in different computer vision tasks, we also summarize the relevant work in different fields.

2.1. FER algorithm based on local features

Although facial detection and cropping have become a standard preprocessing procedure in facial expression recognition, using cropped faces as input to the network increases computational complexity and easily introduces background noise. Since the fact that different facial expressions are generated by muscle movements in multiple regions of the face, some facial local regions associated with facial muscles can be exploited for FER. The facial appearance and geometric features are extracted from facial domain-specific local regions, and all features are fed into a support vector machine for FER (Ghimire et al., 2017). Kumar et al. (2016) proposed an informative region extraction model to obtain important regions based on the projection analysis mechanism. Jan et al. (2018) used facial landmarks to crop four facial parts, i.e., eyebrows, eyes, mouth and nose for FER. Fan et al. (2018) designed

an ensemble deep framework to extract different region features. Hazourli et al. (2020) exploited seven facial patches to construct a patches aggregation framework for FER. Although these models use local facial information in the process of facial expression analysis, they do not consider the close relationship between facial muscle movements and expressions. In this work, we crop facial ROIs associated with facial AUs for facial expressions.

2.2. Lightweight deep models for FER

With the widespread deployment of facial recognition technology in various mobile devices, the deployment demand for facial emotion analysis systems is also increasing. Lightweight models have significant advantages in mobile deployment, and as a result, more and more lightweight networks are designed for expression analysis tasks. Ding et al. (2017) designed a small FER model named FaceNet2ExpNet which only had 11M model parameters. Then, Ding et al. (2017) proposed the PPDN network to reduce the parameters of the model to 6M. Moreover, Kuo et al. (2018b) proposed a compressed facial expression recognition model with model parameters reduced to 1M. Although the parameters of the model have been significantly reduced, the parameters of these models are still at the million level. Recently, Cheng et al. (2020) designed a novel graph convolution network with 100K model parameters. A lightweight deep model which only had 65K model parameters was proposed for frontal face expression analysis (Cugu et al., 2019). But all the above lightweight deep networks did not effectively balance accuracy, computation complexity and speed. In this paper, we design a novel lightweight deep network to achieve higher accuracy than other lightweight networks.

2.3. Vision transformer

The traditional convolution operation can extract local features, which is limited by the local Receptive field and has a weak ability to model sequences. Vaswani et al. (2017) proposed the initial transformer architecture based on a multi-head self-attention mechanism (MSA) which can model long-range dependencies to guarantee good performance in sequence signal processing. The initial transformer architecture has shown great success in natural language processing (Sutskever et al., 2014). Recently, the transformer architecture has been extended to computer vision, i.e., classification (Touvron et al., 2021), object tracking (Chen et al., 2021), segmentation (Cheng et al., 2021) and so on. A series of variant transformers have been proposed, such as swin transformer (Liu et al., 2021), BoTNet (Srinivas et al., 2021), PVT (Wang et al., 2021) and 3DETR (Misra et al., 2021). These methods have achieved performance breakthroughs in many tasks. In FER, different facial expressions relate to different facial muscle movements, and the dependencies of muscles are not local-range. So the transformer architecture has natural advantages to model long-range relations of facial muscles. Inspired by such an idea, we expect the performance of the deep network to be improved by combining the vision transformer and convolution operation in the FER task.

2.4. Binary deep neural network

The binary neural network has achieved great success on resource-limited devices. In the pioneering studies, Hubara et al. (2016) introduced a method to train binarized neural networks (BNNs). Rastegari et al. (2016) designed the XNOR-Net model to prove the effectiveness of the binarization. Liu et al. (2018) introduced shortcuts in the binary network, in which representational capability was significantly enhanced. Zhu et al. (2019) proposed the binary ensemble neural network (BENN) in which the ensemble method was designed to improve the performance. The additional cost of computation was negligible. In the FER task, Sun et al. (2017) proposed binarized auto-encoders and stacked binarized auto-encoders which trained on a large-scale unlabeled facial dataset. The performance can be effectively improved by the transferring mechanism. The whole model was relatively complex, and these models adopt the whole face or its features as the input. Thus, we need to design an efficient lightweight binarized deep network which uses facial local information.

3. The proposed method

3.1. Motivation

Effectively exploring the intrinsic connection between facial action units and emotions can effectively improve the flexibility of emotion recognition. The existing convolutional networks can expand the receptive field of feature extraction by stacking multiple convolutional layers, but it will increase the complexity of the model. Therefore, introducing a multi-head self-attention mechanism can achieve modeling of long-distance dependencies between different facial units. In addition, efficient binary operations can effectively reduce the memory requirements and improve the inference speed. In this section, we design three lightweight deep networks: Mini-baseline, MiniTR and MiniBTR. Our proposed Mini-baseline only includes two 1D convolution layers and fully connected layers. To model long-range dependencies of facial muscle movements, we introduce the transformer encoder at the end of the last convolution layer in the Mini-baseline. The transformer encoder has the advantage of exploiting the relationship among image patches. In our proposed MiniTR which is shown in Fig. 2, we only adopt the AU associated regions to construct these image patches. As shown in Fig. 4, our MiniBTR is a variant of MiniTR by introducing the binary operation for effectively deploying the model on edge devices.

3.2. Mini-baseline network designing

Although neural architecture search (NAS) (Liu et al., 2019) has played an increasingly critical role in network designing, a large amount of computation is still needed to search for network structures that meet constraint conditions in the search space. In this paper, we adopt the human–machine collaborative strategy (Lee et al., 2020) to design a deep network model. The whole procedure has two stages: principled network design by humans and machine-driven design.

Firstly, a principled network architecture ψ is designed by human experiments. According to the conclusion in Khorrami et al. (2015) and Cugu et al. (2019), when the depth of the deep network is four or five, the deep model achieves comparable results for frontal face expression analysis. As the input of the deep model is the HOG feature and the dimension of the HOG feature is 144, we adopt two 1D convolution layers for feature extraction. As the fully connected layer is widely used in classification tasks, we utilize two fully connected layers. Secondly, we use the machine-driven design mechanism to decide the configuration of deep network architecture. The whole procedure is based on the generative synthesis (Wong et al., 2018) algorithm which forms the search for the optimal network structure into a constrained optimization problem. Specifically, the generative synthesis algorithm finds a generator $\mathcal{P}$ which adopts a set of seeds $\mathcal{S}$ to generate networks $\{\mathcal{N}_s | s \in \mathcal{S}\}$. Meanwhile, we introduce a universal function $\mathcal{M}$ to measure the performance of network design based on indicator function $1_r(\cdot)$. In this paper, we utilize the NetScore (Wong, 2019) to verify the effectiveness of network design. The optimization problem of the generative synthesis algorithm is defined as follows:

$$\mathcal{P} = \arg\max_{\mathcal{P}} \mathcal{M}(\mathcal{P}(s)) \ subject\ to\ 1_r(\mathcal{P}(s)) = 1,\ for\ all\ s \in \mathcal{S}, \tag{1}$$

where the NetScore (denoted here as $\mathcal{M}$) is calculated as:

$$\mathcal{M}(\mathcal{N}) = 20\log\left(\frac{e(\mathcal{N})^{\eta}}{p(\mathcal{N})^{\mu} m(\mathcal{N})^{\nu}}\right) \tag{2}$$

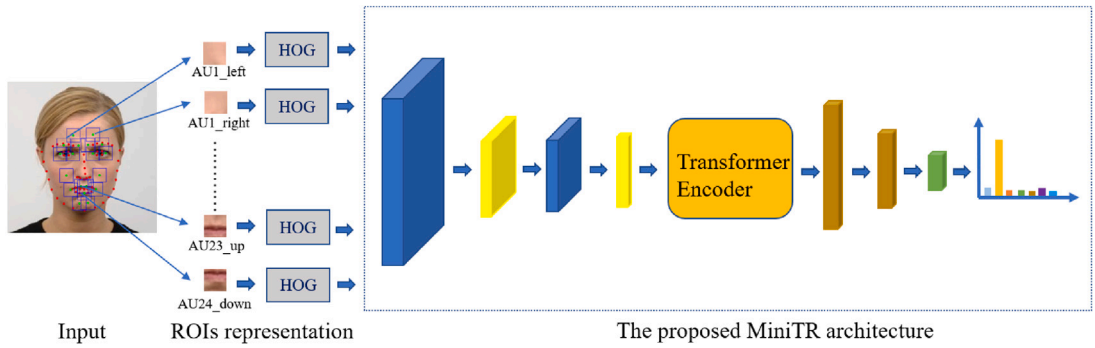


Fig. 2. An overview of the proposed deep neural network of MiniTR. Compared to our proposed Mini-baseline, we add the transformer encoder at the last 1D convolution layer. By introducing 4 heads in MSA mechanism. The parameters of MiniTR are 69K.

Table 1
The configuration of MiniTR based on HOG feature.

Layer	Layer configuration	Number of parameters
Input	(20, 144)	0
Convolution	conv1d(144, 64)	144 × 64 + 64
Convolution	conv1d(64, 32)	64 × 32 + 32
Transformer encoder	4 × 3 × conv1d(32, 32)	4 × 3 × 1024
Flatten layer	640	0
Fully connected layer	640	640 × 64 + 64
Fully connected layer	7	64 × 7 + 7
Softmax normalization layer	7	0

where $\mathcal{N}$ denote the generated deep network, $e(\mathcal{N})^{\eta}$ and $p(\mathcal{N})^{\mu}$ are the accuracy and the parameters of $\mathcal{N}$, respectively. $m(\mathcal{N})^{\nu}$ is the number of multiply-accumulate operations during network inference, and η, μ, ν are key coefficients to control the weight of different indicators. In this paper, we set $\eta = 2, \mu = 0.5, \nu = 0.5$ for deep neural measuring.

We adopt stricter indicator functions to ensure the efficiency of network design. Compared to the reference paper (Lee et al., 2020), We set the constraint conditions as: (1) the recognition accuracy on the CK+ dataset is higher than 93% (Lucey et al., 2010), and (2) the total number of parameters in the model is less than 1M.

3.3. MiniTR network designing

Although the Mini-baseline is an efficient lightweight model which only uses traditional convolution operation, the relationships among different muscle movements are not effectively considered. As facial muscle movements are long-range in different facial expressions, the transformer architecture which exploits the relationship among image patches can be seamlessly incorporated into Mini-baseline to model long-range muscle movements. Based on those findings in Raghu et al. (2021) and Wu et al. (2021), the following design rules of MSA are given as follows: (1) replacing the Convs with MSA blocks from the end of a baseline CNN model, (2) replacing the ahead Convs with MSA blocks if rule (1) does not obtain good performance, (3) using more heads and higher hidden dimensions for MSA, we propose a novel framework MiniTR by adding the transformer encoder in Mini-baseline (see Fig. 2).

The transformer encoder includes three key parts: convolutional token embedded, MSA and residual connection, where the input of the transformer is N different query tokens $q_1, q_2, \ldots, q_N$. In our proposed framework, all embedding patches are constructed based on the deep feature of image patches which pass two 1D convolution layers. We denote all tokens as $X_q \in \mathbb{R}^{N \times L}$, where L is the dimension of deep features. In this paper, we set L as 32. The query Q, key K and value V which are projected from X_q according to Eq. (3).

$$Q = X_q W_Q, K = X_q W_K, V = X_q W_V \tag{3}$$

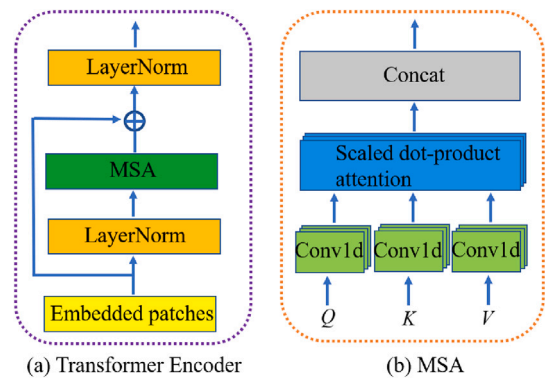


Fig. 3. Illustration of transformer encoder. (a) shows the whole transformer encoder. (b) shows the MSA module. We adopt 1D convolution to project input into embedded space rather than linear transformation.

where W_Q, W_K, W_V are learnable projection matrices. In our transformer encoder, we adopt the 1D convolution operation to produce corresponding Q, K and V rather than linear projection. By using 1D dimensional convolution, the local facial muscle motion relationships can be described through the local receptive field in convolution operation. So the Q, K and V are calculated as follows:

$$Q = conv1d_q(X_q), K = conv1d_k(X_q), V = conv1d_k(X_q) \tag{4}$$

where $\mathrm{conv1d_q}$, $\mathrm{conv1d_k}$ and $\mathrm{conv1d_v}$ denote three 1D convolution layers. The self-attention operation of each head is calculated in Eq. (5):

$$Att_i(Q_i, K_i, V_i) = \mathrm{softmax}(\frac{Q_i K_i^T}{\sqrt{d}})V_i \tag{5}$$

where Q_i, K_i and V_i are query, key and value in the ith head, respectively. Att_i is the result from the i head. The H is the number of attention heads. The MSA is defined in Eq (6):

$$MSA(Q, K, V) = \mathrm{concat}[\mathrm{Att_i(Q_i, K_i, V_i)}]_{\mathrm{h=1...H}}\mathrm{W_o} \tag{6}$$

where concat is a concatenation operation based on H head features, W_o is a learnable parameter matrix. At last, the output of the transformer adopts the residual connection, which is calculated as follows:

$$X_{out} = X_p + MSA(Q, K, V) \tag{7}$$

Specifically, in our proposed MiniTR framework, we use four heads in the transformer encoder, and the output dimension of each MSA is 32. The transformer encoder is shown in Fig. 3. The parameters of the MiniTR are 69K (see Table 1). By exploiting the transformer encoder, the long-range relationships of different tokens which relate to different AUs can be effectively modeled, and the performance can be further improved in FER.

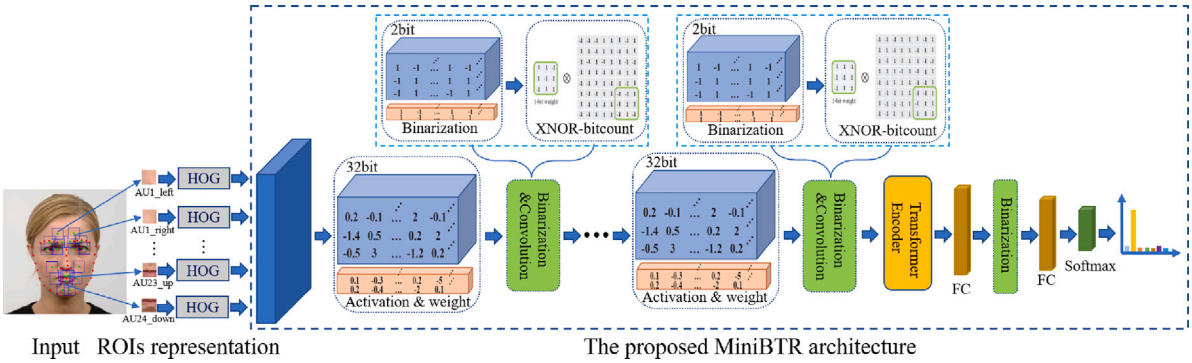


Fig. 4. An overview of the proposed deep neural network of MiniBTR. We add binarization and XNOR-bitcount operation into the MiniTBR architecture. By introducing the binary quantization mechanism, we can effectively reduce computational cost and memory resources. During the quantization error, the model suffers a decrease in accuracy even introducing scale factors.

3.4. MiniBTR network designing

To reduce the memory and computing power of the MiniTR, we introduce the low-bit quantization mechanism. Recently, low-bit quantization has been widely used for deploying deep models on mobile devices, in which binarized neural networks have gained substantial interest. By exploiting the binarization function, all the weight and activation values of BNNs are composed of only −1 and +1. The sign function can be utilized for the binarization function, which is defined as follows:

$$\text{sign}(x) = \begin{cases} +1, & \text{if } x \geq 0, \\ -1, & \text{otherwise}, \end{cases} \tag{8}$$

where x is the input.

The full precision (32-bit floating point) weight $\mathcal{W} \in \mathbb{R}^{c\times w\times h}$ and activation $\mathcal{A} \in \mathbb{R}^{c\times w\times h}$ are binarized to −1 or +1 by binarization function, which is defined as follows:

$$Q_b(\mathcal{W}) = sign(\mathcal{W}), Q_b(\mathcal{A}) = sign(\mathcal{A}) \tag{9}$$

where $Q_b(\mathcal{W})$ and $Q_b(\mathcal{A})$ are binary weight and binary activation.

BNNs with binary weights and activations drastically reduce memory size. Furthermore, the 32-bit floating-point multiply-accumulations can be replaced by 1-bit XNOR-Bitcount operations, which can lead to a great increase in power efficiency. This could have a big impact on dedicated deep-learning hardware. The convolution operation of the forward process of the network is defined as follows:

$$\mathcal{Z} = \sigma(Q_b(\mathcal{W}) \otimes Q_b(\mathcal{A})) = \sigma(Q_b(\mathcal{W}) \odot Q_b(\mathcal{A})) \tag{10}$$

where σ is the activation function, $\mathcal{Z}$ is the output, $\otimes$ is the traditional convolution operation, $\odot$ is the XNOR-bitcount operations.

If the weight and activation of the deep network are processed by the $sign(\cdot)$ function directly, this binarized mechanism introduces a high quantization error and leads to low accuracy. Our goal is to design an efficient binarized model that reduces computational costs while preserving accuracy. To further reduce the quantization error, Rastegari et al. (2016) proposed XNOR-net to introduce the scaling factor into the network weight and activation quantization process while binarizing. Let $\boldsymbol{w}, \boldsymbol{a} \in \mathbb{R}^n, n = c \times w \times h$ denote the vector of weight and activation. The quantization procedure of weight and activation including the scale factor is calculated as follows:

$$Q_b(\boldsymbol{w}) = \alpha \boldsymbol{b}, Q_b(\boldsymbol{a}) = \beta \boldsymbol{h} \tag{11}$$

where $\boldsymbol{b}$, $\boldsymbol{h}$ are binary weight and activation vector, respectively. $\boldsymbol{b} \in \{+1, -1\}^n, \boldsymbol{b}^T\boldsymbol{b} = n$, $\boldsymbol{h} \in \{+1, -1\}^n$, α and β are scaling factors.

To approximate the dot product of $\boldsymbol{a}$ and $\boldsymbol{w}$, $\boldsymbol{a}^T\boldsymbol{w} \approx \beta\boldsymbol{h}\alpha\boldsymbol{b}$, the optimization problems are solved as follows:

$$\alpha^*, \boldsymbol{b}^*, \beta^*, \boldsymbol{h}^* = \underset{\alpha,\boldsymbol{b},\beta,\boldsymbol{h}}{\arg\min} \|\boldsymbol{a} \odot \boldsymbol{w} - \alpha\beta\boldsymbol{h}\boldsymbol{b}\|^2 \tag{12}$$

Let $y_i = a_i w_i, c_i = h_i(b)_i, \gamma = \alpha\beta$, the above optimization problems are further transformed into the following formula:

$$\gamma^*, \boldsymbol{c}^* = \underset{\gamma,C}{\arg\min} \|\boldsymbol{y} - \gamma\boldsymbol{c}\| \tag{13}$$

where $\boldsymbol{c}^* = Sign(\boldsymbol{a}) \odot Sign(\boldsymbol{w}) = \boldsymbol{h}^* \odot \boldsymbol{b}^*$, $\gamma^* = \frac{\sum Y_i}{n} \approx (\frac{1}{n}\|\boldsymbol{a}\|_{l_1})(\frac{1}{n}\|\boldsymbol{w}\|_{l_1}) = \beta^*\alpha^*$.

By introducing scale factors, we design a novel binarized deep network by extending the proposed MiniTR. The proposed binarized deep network MiniBTR for FER is shown in Fig. 4. After each 1D convolution layer and the second to last fully connected layer, the binarization and XNOR-Bitcount operation layer is embedded. We conduct extensive comparative experiments to validate the effectiveness of the Mini-baseline, Mini-one-head (we introduce one head in MSA for comparative analysis), MiniTR (we adopt four heads in MSA) and MiniBTR.

4. Experiment

Firstly, we introduce the data preprocessing details and training settings. Next, we compare our proposed models with existing lightweight deep networks. Finally, we compare different feature representations and network design.

4.1. Facial expression datasets and preprocessing

We conduct experiments on four facial expression datasets, i.e., Radboud Faces Database (RaFD) (Langner et al., 2010), Extended Cohn-Kanade (CK+), Oulu-CASIA (Zhao et al., 2011) and RAF-DB (Li et al., 2017). The first three datasets are laboratory-controlled facial expression datasets, and the samples of the last dataset are collected from the Internet. There are 67 subjects on RaFD dataset, and each subject has eight types of facial expressions. In our experiment, we only utilize frontal face images with some types of expressions, i.e., anger, disgust, fear, happiness, sadness, surprise and neutral. The total number of samples is 1407 (67 subjects × 7 expressions × 3 gaze directions). The picture size of each example is 681 × 1024. On CK+ and Oulu-CASIA datasets, we use 327 and 480 video sequences. In each video sequence, only the last three frames are adopted. On RAF-DB dataset, we adopt the basic emotion dataset which includes 15,339 images. We use 12,271 images for training and samples and 3068 images for testing. Some samples from four facial expression datasets are shown in Fig. 5.

As facial expressions have a close relation to different AUs, the combinations of different AUs produce corresponding basic facial expressions. Cropping AU-associated facial regions has two-fold advantages. On the one hand, the background noise and the size of the input can be reduced. On the other hand, the performance can be further

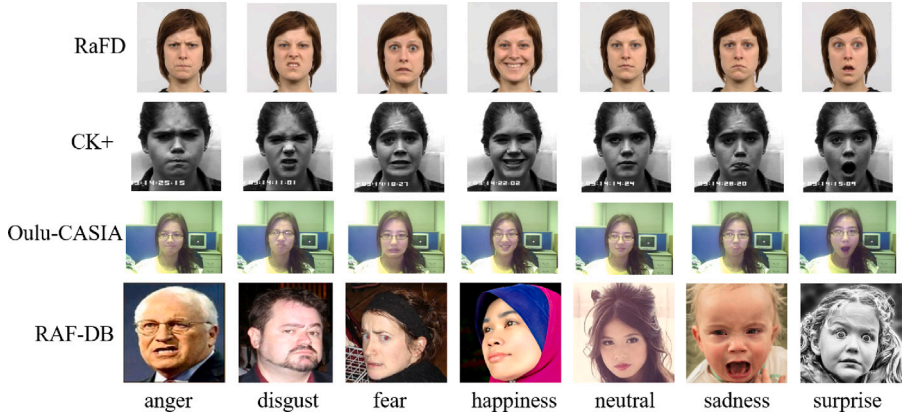


Fig. 5. Some samples of RaFD, CK+, Oulu-CASIA and RAF-DB datasets.

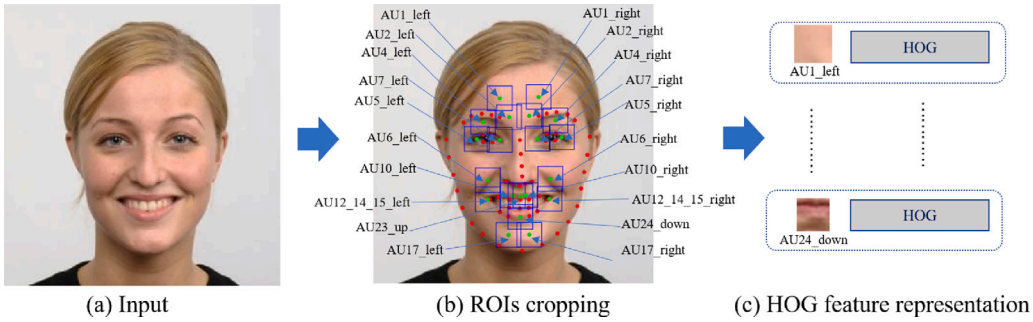


Fig. 6. Facial ROIs cropping and representation.

improved by extracting features from specific regions. We aim to crop AU-associated facial local regions. We locate specific ROI centers by facial landmarks according to existing research (Jin et al., 2021). We work on 20 AUs in the experiments. The whole cropping procedure is shown in Fig. 6. Firstly, we exploit some facial landmarks (the red spots in Fig. 6(b)) to crop ROIs. All centers of ROI are shown in green spots in Fig. 6(b). After cropping facial local regions which are shown in Fig. 6(c), we represent these ROIs by the HOG feature which can reduce the computational cost and ensure higher accuracy. The HOG features have a stronger ability to describe the subtle changes of different muscle textures. Some settings of different datasets are similar to the following references: RaFD with Ferro-Pérez and Mitre-Hernandez (2020), CK+ with Kuo et al. (2018), Liu et al. (2014), Oulu-CASIA with Cugu et al. (2019), respectively. For our training and testing, we used a desktop computer with an Intel Core i74790 CPU with 32 GB memory and an NVIDIA GeForce GTX 1070 graphics card with 6 GB memory. The algorithm code was created in Python 3.6 and Pytorch 1.5. All deep models are run on Anaconda 3.0 software and debugged in Pycharm IDE. We set the batch size of 21 or 18 for 6 and 7 kinds of emotions. The initial learning rate is 0.001 and the decay step is set to 50, and we adopt the maximum iteration as 1000. We used cross-validation to measure the recognition accuracy of facial expression recognition on RaFD, CK+, Oulu and CASIA datasets. We performed five-fold cross-validation on the RaFD and Oulu-CASIA datasets refer to Liu et al. (2023) and Hussein et al. (2022) . We divided the two datasets into five folds, where four-fold samples are used for model training and one-fold samples are used for model testing, respectively. On the CK+ dataset, we used 10 sets of cross-validation. We divide all the subjects of the CK+ dataset into 10 subsets according to the same settings refer to Kuo et al. (2018b). While on RAF-DB dataset, we calculate the average accuracy of facial expression recognition in five experiments on the test set.

4.2. Results on four facial expression datasets

To evaluate the proposed models, we have carried out a lot of comparative experiments. Because we mainly design lightweight networks, we adopt existing lightweight networks in our comparison experiment. When performing comprehensive performance analysis, we focus on the accuracy, size and speed of different models.

As seen in Table 2, IDNN (Mollahosseini et al., 2016) achieves 92% accuracy by using the most model parameters. Particularly, the MobileNet (Ioffe and Szegedy, 2015) also obtains the worst results. Although DDAL (Jin et al., 2020) and Attention (Sun et al., 2018), get better performances, these methods exploit deeper networks or special mechanisms, i.e., labeled and unlabeled data are utilized simultaneously in DDAL, and a weight mask is learned automatically in Attention model. MicroExpNet (Cugu et al., 2019) which does not adopt the distillation learning mechanism obtains 90.08% accuracy by using a smaller model. The proposed Mini-baseline obtains a 93.92% recognition rate with 52K parameters. Compared to MicroExpNet, the accuracy increases by 3.12% by using facial local information. By introducing a one-head self-attention mechanism, Mini-one-head outperforms Mini-baseline by 0.46%. While increasing the number of heads, the performance can be further improved. In MiniTR, there are four heads in MSA, and we obtain the highest accuracy 95.94% with only 69K parameters. In short, the proposed MiniTR achieves exciting results compared to existing lightweight networks on RaFD dataset.

Table 3 reports the results of the proposed deep network and the compared methods on CK+ dataset. By using deep dynamic features, the performance can be effectively improved in some models, such as DTAN (Jung et al., 2015), Peak-Piloted (Zhao et al., 2016) and CDLM-frame2seq (Kuo et al., 2018). The Peak-Piloted obtains 99.30% accuracy. Moreover, for CDLM-frame2seq (Kuo et al., 2018), the accuracy drops to 98.47%. Both above models adopt dynamic features. However, some efficient deep networks are proposed based on static

Table 2
Comparisons of different methods on RaFD dataset.

Method	Accuracy (%)	Parameters
IDNN (Mollahosseini et al., 2016)	92.00	16 158K
EDNN (Jain et al., 2019)	95.00	4621K
1.0 MobileNet (Ioffe and Szegedy, 2015)	55.00	3235K
0.75 MobileNet (Ioffe and Szegedy, 2015)	48.00	1837K
PeleeNet (Wang et al., 2018)	84.00	2123K
ResMoNET (Ferro-Pérez and Mitre-Hernandez, 2020)	90.00	1721K
DDAL (Jin et al., 2020)	94.63	343K
Attention (Sun et al., 2018)	97.20	369K
L3Net (Cheng et al., 2020)	93.25	102K
DDRGCN (Jin et al., 2021)	94.48	110K
MicroExpNet (Cugu et al., 2019)	90.08	65K
MobileViT (Jiang et al., 2024)	92.30	972K
MobileViT-XXS (Wang and Zhang, 2023)	95.45	1298K
Improved MobileNetV3 (Jiang et al., 2024)	95.80	2560K
SNNEs (Altaf et al., 2025)	90.00	–
Mini-baseline	93.92	52K
Mini-one-head	94.38	55K
MiniTR	95.94	69K

Table 3
Comparisons of different methods on CK+ dataset.

Method	Feature	Accuracy (%)	Parameters
DTAN (Jung et al., 2015)	Dynamic	97.25	5950K
Peak-Piloted (Zhao et al., 2016)	Dynamic	99.30	6798K
CDLM-frame2seq (Kuo et al., 2018)	Dynamic	98.47	2690K
BASIC-FRENET (Tang et al., 2020)	Static	98.97	22 528K
BLOCK-FRENET (Tang et al., 2020)	Static	98.91	9216K
CDLM-frame (Kuo et al., 2018)	Static	97.37	2673K
CDLM-tiny (Kuo et al., 2018)	Static	96.81	1229K
EmotionNet Nano-A (Lee et al., 2020)	Static	97.60	232K
EmotionNet Nano-B (Lee et al., 2020)	Static	92.70	136K
L3Net (Cheng et al., 2020)	Static	94.12	102K
DDRGCN (Jin et al., 2021)	Static	94.32	110K
MicroExpNet (Cugu et al., 2019)	Static	90.71	65K
Rs-xception (Liao et al., 2024)	Static	97.13	1.92M
LITE-FER (Bicer and Kose, 2024)	Static	89.69	113K
LCNN (Grover and Bansal, 2025)	Static	98.0	1331K
SLW-LDT (Chen et al., 2025b)	Static	98.9	1330K
Mini-baseline	Static	93.40	52K
Mini-one-head	Static	93.87	55K
MiniTR	Static	94.56	69K

features. In BASIC-FRENET (Tang et al., 2020), the accuracy is 98.97% by only using static features. However, the BASIC-FRENET still has too many parameters. Recently, some lightweight models have been designed. EmotionNet Nano-A (Lee et al., 2020) obtains a 97.6% recognition rate with 232K parameters. Moreover, the simplified version EmotionNet Nano-B (Lee et al., 2020) only has 136K parameters. In Mini-baseline, we get an accuracy of 93.40% with 52K parameters. Compared to MicroExpNet, the recognition rate increases by 2.69% by using facial local information in Mini-baseline. The background noise decreases the performance by using the whole face as the input in MicroExpNet. Moreover, we obtain 94.56% and 93.87% by introducing the self-attention mechanism in Mini-one-head and MiniTR. Interestingly, the results of our proposed models show that using only the static feature and lightweight models can still achieve good results in MiniTR and Mini-one-head.

In Table 4, using dynamic features can achieve better performance, e.g., DTAN (Jung et al., 2015) and CDLM-frame2seq (Kuo et al., 2018) on Oulu-CASIA dataset. The CDLM-frame2seq (Kuo et al., 2018) achieves the highest accuracy at 91.67%. Meanwhile, some models which use static features still obtain comparable results. The accuracy of MicroExpNet (Cugu et al., 2019) is 56.71% by using static features. Although MicroExpNet utilizes a lightweight model, the performance is undesirable. Compared to MicroExpNet (Cugu et al., 2019), L3Net (Cheng et al., 2020) gets better performance by exploiting the graph convolution network in which facial local regions were exploited. Compared to L3Net (Cheng et al., 2020), the performance is increased by 8.12% in DDRGCN by exploiting dynamic relationships among AU-based ROIs. In our proposed models, we achieve exciting results with a smaller model. In Mini-baseline, we obtain 73.26% accuracy with 52K parameters. We boost performance by introducing the self-attention mechanism in Mini-one-head and MiniTR. Although this dataset has many challenges in expression recognition tasks, the proposed model MiniTR still obtains 76.10% accuracy.

As shown in Table 5, we report the results of different deep models on RAF-DB dataset. DLP-CNN+mSVM (Li and Deng, 2018) gets the best performance with 74.38% accuracy and 19 655K parameters. This dataset is collected from facial emotion images in real environments. Due to various challenges in complex scenes, such as lighting and head posture, the accuracy of lightweight model recognition is relatively low. The lightweight deep model MicroExpNet obtains the lowest accuracy. The accuracies of L3Net (Cheng et al., 2020) and DDRGCN (Jin et al., 2021) are 55.03% and 58.25%, respectively. The L3Net (Cheng et al., 2020) are 9.72% and 12.94% better than MicroExpNet (Cugu et al., 2019) by using facial local regions. However, LWFER achieved the highest recognition accuracy using only 770K model parameters. In our models, Our proposed model obtains the HOG features associated with AUs through facial landmark information, effectively alleviating the interference of background noise. In addition, by introducing a multi-head self-attention mechanism, we can exploit the long-distance dependencies of AUs, effectively improving the performance of emotion recognition. We achieve 61.22%, 61.86% and 62.45% accuracies, respectively. Compared to DDRGCN, we achieve better performance with fewer parameters. However, in complex scenarios, deviations in

Table 4
Comparisons of different methods on Oulu-CASIA dataset.

Method	Feature	Accuracy (%)	Parameters
DTAN (Jung et al., 2015)	Dynamic	74.38	5950K
CDLM-frame2seq (Kuo et al., 2018)	Dynamic	91.67	2690K
CDLM-frame (Kuo et al., 2018)	Static	88.75	2673K
CDLM-Tiny (Kuo et al., 2018)	Static	85.84	1229K
DDAL (Jin et al., 2020)	Static	61.87	343K
STSGN (Zhou et al., 2020)	Dynamic	87.23	337K
ASModel (Sun et al., 2021)	Dynamic	70.83	300K
WAM (Jingying et al., 2024)	Dynamic	73.40	10 127K
LBAGCN (Jingying et al., 2024)	Dynamic	87.71	5611K
L3Net (Cheng et al., 2020)	Static	65.16	102K
DDRGCN (Jin et al., 2021)	Static	73.28	110K
MicroExpNet (Cugu et al., 2019)	Static	56.71	65K
TinyMLFER (Bai and Yu, 2025)	66.02	70K	
Mini-baseline	Static	73.26	52K
Mini-one-head	Static	74.48	55K
MiniTR	Static	76.10	69K

Table 5
Comparisons of different methods on RAF-DB dataset.

Method	Accuracy (%)	Parameters
DLP-CNN+mSVM (Li and Deng, 2018)	74.38	19 655K
FSN (Zhao et al., 2018)	72.46	60 966K
CDLM-frame (Kuo et al., 2018)	65.52	2673K
L3Net (Cheng et al., 2020)	55.03	102K
DDRGCN (Jin et al., 2021)	58.25	110K
MicroExpNet (Cugu et al., 2019)	45.31	65K
MobileViT-XXS (Wang and Zhang, 2023)	73.89	1298K
LWFER (Zeng et al., 2023)	86.92	770K
CLCM (Gursesli et al., 2024)	84.0	2393K
AFNet (Chen et al., 2025a)	80.3	26 480K
Mini-baseline	61.22	52K
Mini-one-head	61.86	55K
MiniTR	62.45	69K

facial landmark information can affect the extraction of local region features. Therefore, our proposed model has more obvious advantages in analyzing facial emotions for frontal faces. At last, we added the confusion matrix of our proposed model on four datasets which is shown in Fig. 7. The corresponding relationship between different abbreviations and different expressions is: A: anger, D: disgust, F: fear, H: happiness, N: neutral, Sa: sadness, Su: surprise. By introducing four confusion matrices, the effectiveness of our model in recognizing different types of facial expressions can be further verified.

However, accuracy and model parameters are only two general indicators to measure the performance of the lightweight deep network design. To validate the effectiveness of different indicators comprehensively, we further conduct more comparative experiments in the following experiments.

4.3. Results of binary deep neural network

We verify the effectiveness of different lightweight binarized deep models in this subsection. We adopt five lightweight deep models, DDRGCN, MicroExpNet and our proposed models. Two kinds of binarization methods are used, such as binarization (*-BNN for short) or binarization with scale factor (*-XNOR for short). Some methods, i.e., MicroExpNet-BNN, MicroExpNet-BNN, Mini-baseline-BNN, and MiniTR-BNN, are binarized directly. Then, MicroExpNet-XNOR, MicroExpNet-XNOR, Mini-baseline-XNOR and MiniBTR introduce scale factors during binarizing. All results are shown in Table 6.

From the results in Table 6, we confirm that the performance is reduced by introducing the quantitative mechanism. However, different quantization mechanisms have a significant impact on model performance. Some models, i.e., DDRGCN-BNN and MicroExpNet-BNN directly quantify weights and activations have a significant decrease in accuracy in facial expression recognition tasks. The accuracies of DDRGCN-BNN and MicroExpNet-BNN are 89.85% and 85.32% on RaFD dataset. To alleviate the impact of quantization errors, the recognition accuracy of the model is improved by introducing a scale factor in the quantization process. The accuracies of DDRGCN- XNOR and MicroExpNet- XNOR are 91.85% and 87.66%, respectively. Therefore, by introducing a quantization mechanism with scale factors, the impact of quantization errors can be effectively alleviated. Compared to full precision operation, using two kinds of binarization methods in different deep models led to significant degradation in accuracy, i.e., the accuracy degradation from 95.05% (MiniTR) to 91.88% (MiniTR-BNN) on RaFD dataset. By introducing scale factors during binarization, we boost performance by 2.26% from 91.88% (MiniTR-BNN) to 94.14% (MiniBTR). Compared to other binary models, MiniBTR model obtains better performance on four facial expression datasets. In summary,

Table 6
Accuracy (%) of different binary deep models.

Models	RaFD	CK+	Oulu-CASIA	RAF-DB
DDRGCN-BNN	89.85	89.73	68.40	47.20
MicroExpNet-BNN	85.32	85.66	50.34	40.21
Mini-baseline-BNN	90.80	90.62	69.09	48.95
Mini-one-head-BNN	91.50	91.34	70.22	49.20
MiniTR-BNN	91.88	91.73	70.45	50.88
DDRGCN-XNOR	91.85	91.67	71.66	50.24
MicroExpNet-XNOR	87.66	87.82	52.36	43.68
Mini-baseline-XNOR	93.14	91.67	71.66	52.60
Mini-one-head-XNOR	93.66	91.88	72.12	53.48
MiniBTR	94.14	92.67	72.66	56.60

	A	D	F	H	N	Sa	Su
A	93.01	1.91	0	0	3.17	1.91	0
D	0	95.88	0	0	2.53	0	1.59
F	0	0	92.39	0	3.17	1.27	3.17
H	0	0	0	100	0	0	0
N	0	0	0	0	100	0	0
Sa	2.53	0	0	0	3.17	92.39	1.91
Su	0	1.91	0	0	0	0	98.09

(a) Confusion matrix on RaFD dataset

	A	D	F	H	Sa	Su
A	93.68	1.58	0	0	2.11	2.63
D	0	94.74	1.05	0	1.05	3.16
F	0	2.11	92.62	0	2.11	3.16
H	0	0	0	100	0	0
Sa	5.26	0	3.69	0	88.42	2.63
Su	0	2.10	0	0	0	97.9

(b) Confusion matrix on CK+ dataset

	A	D	F	H	N	Sa	Su
A	77.07	10.07	4.17	1.74	1.74	3.47	1.74
D	8.68	70.83	6.25	2.08	2.78	5.21	4.17
F	6.25	5.21	74.65	2.43	4.17	2.78	4.51
H	3.13	1.74	1.74	85.07	1.74	3.47	3.11
N	4.51	3.74	4.17	1.74	76.46	4.17	5.21
Sa	6.25	3.82	6.59	1.75	5.21	74.58	2.43
Su	1.91	6.94	5.91	4.48	4.17	2.28	74.31

(c) Confusion matrix on Oulu-CASIA dataset

	A	D	F	H	N	Sa	Su
A	60.45	8.38	9.54	4.36	6.63	6.23	4.41
D	7.68	55.22	8.25	6.32	5.28	8.65	8.60
F	11.03	6.45	55.04	4.78	6.86	8.22	7.62
H	2.25	3.64	4.86	76.51	3.42	4.36	4.96
N	6.65	5.96	7.25	3.68	66.66	6.62	3.18
Sa	7.56	4.85	7.56	4.33	7.05	61.35	7.30
Su	5.36	8.25	5.28	4.26	5.65	8.75	62.05

(d) Confusion matrix on RAF-DB dataset

Fig. 7. confusion matrix of our proposed model on four datasets.

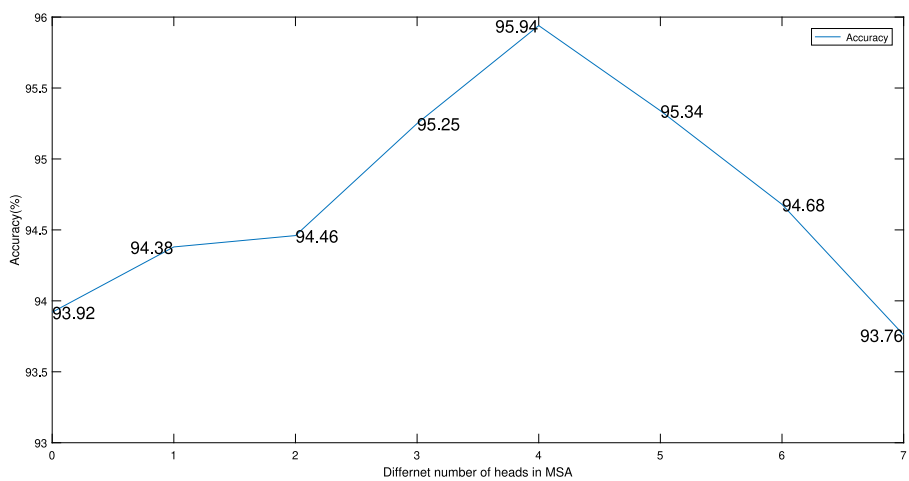


Fig. 8. Accuracy of different head numbers in MSA on RaFD dataset.

our proposed efficient and lightweight two-level model is used for sentiment analysis. This model fully exploits the local HOG features of facial action units to alleviate the influence of factors unrelated to facial expressions, such as background noise and lighting changes. In addition, we achieved modeling of long-distance facial muscle movements by constructing a hybrid backbone network based on CNN and Transformer. Finally, we improve the derivation speed of the model by introducing a scale factor-based quantization mechanism without significantly reducing recognition accuracy.

4.4. Effects of head numbers in MSA

In the FER task, the performance can be improved by introducing the MSA mechanism. Although increasing the number of heads in MSA has a positive effect on the improvement of performance, the parameters of the model also increase remarkably. Especially in the lightweight model design, we need to compromise model parameters and accuracy. So in this experiment, we further validate the effect of head numbers in MSA. We conduct a 5-fold cross-validation on RaFD dataset, and we set the number of the head between 0 and 7 with 1 interval.

We do not introduce the self-attention mechanism in Mini-baseline. In Mini-one-head, the number of heads is set as 1. As all results are shown in Fig. 8, by increasing the number of heads, the accuracy can be improved significantly from 94.38% (Mini-one-head) to 95.94% (MiniTR which has 4 heads in MSA). When the number of heads is equal to 4, we obtain the best performance. The accuracy reduces if we keep on increasing the number of heads. So in our proposed model MiniTR, the number of heads is set as 4.

4.5. Model size analysis

The size of the model is an important indicator for measuring network mobility deployment. The model size is calculated by MByte from three aspects: input size, forward/backward pass size and parameter size. This indicator is mainly used to count the memory usage of the model. We compare some lightweight deep models, such as FN2EN (Ding et al., 2017), PPDN (Zhao et al., 2016), and lightweight models, such as DDRGCN (Jin et al., 2021), L3Net (Cheng et al., 2020), MicroExpNet (Cugu et al., 2019), some variant StudentExpNet (Cugu et al., 2019) and our proposed models.

As the results are shown in Table 7, The model size of Mini-baseline is only 0.23 MB, which is the smallest model. Compared to

Table 7
Memory size of different models.

Model	Size (MByte)
FN2EN (Ding et al., 2017)	35.32
PPDN (Zhao et al., 2016)	50.13
StudentExpNet_M (Cugu et al., 2019)	12.08
StudentExpNet_S (Cugu et al., 2019)	3.17
StudentExpNet_XS (Cugu et al., 2019)	1.67
MicroExpNet (Cugu et al., 2019)	0.93
L3Net (Cheng et al., 2020)	0.59
DDRGCN (Jin et al., 2021)	0.48
Mini-baseline	0.23
Mini-one-head	0.26
MiniTR	0.36

the DDRGCN, the Mini-baseline only needs half of its memory space. However, Mini-baseline does not show the best accuracy. So in MiniTR, we introduce the transformer encoder to improve the performance. Compared to DDRGCN, the MiniTR achieves higher accuracy with a smaller model size.

4.6. Model speed analysis

In this subsection, we compare the inference speed of some light weight deep models, such as MicroExpNet (Cugu et al., 2019), and variant StudentExpNet (Cugu et al., 2019). We calculate the average time of validation set on RaFD dataset during the forward procedure. In our experiments, we use two kinds of hardware platforms: GPU (GTX1070) and CPU (I7_4790).

Table 8 reports the average time of different models. We compared the inference speed of different models on two hardware platforms. These lightweight models all use full precision weights and activation values. The MicroExpNet which has the fewest model parameters, obtains 5.61 ms and 1.11 ms inference speed on two hardware platforms. From the perspective of actual deployment, the real-time performance of MicroExpNet has already met practical needs. Our proposed MiniTR has a faster inference speed than MicroExpNet. Although Mini-baseline and Mini-one-head achieve better performance compared to MiniTR, the speed difference among the three models on different hardware platforms is relatively small. And after binarization, the speed of inference will be greatly improved. Although quantization errors can affect the accuracy of emotion recognition, introducing a scale factor allows the model to achieve a good compromise between accuracy and speed.

4.7. Effects of different feature representation for ROIs

Although we utilize the HOG feature to represent each ROI for FER, the performance could be different by using different feature representations. We adopt three traditional hand-crafted features, such as scale-invariant feature transform (SIFT), histogram local binary pattern (HLBP), HOG, and deep feature from deep auto-encoder (DAE). The dimensions of SIFT and HLBP are 128 and 256, respectively. DAE can The DAE can realize an effective compressed representation of features. We use three abbreviations, i.e., AE32, AE64 and AE128, to represent the different dimension features from the hidden layers of DAE, and the number denotes the dimension of the deep feature.

All the results are shown in Fig. 9. In all traditional hand-crafted features, the HOG feature reaches the highest accuracy at 95.94%. With the dimensions of AE- increase, the recognition accuracy improves significantly. When using the feature of AE128, we get the best performance (96.24%). Although the deep features can obtain higher accuracy, the complexity of the system and preprocess increases. To obtain the deep features, the DAE is pre-trained. In each forward procedure, the ROIs pass the DAE to produce the deep features. Model accuracy is the sole indicator to evaluate deep neural networks. Computational complexity, memory size and speed are other important indicators, especially for the deep model deploying on edge devices. For the proposed lightweight deep models, we focus on comprehensive performance.

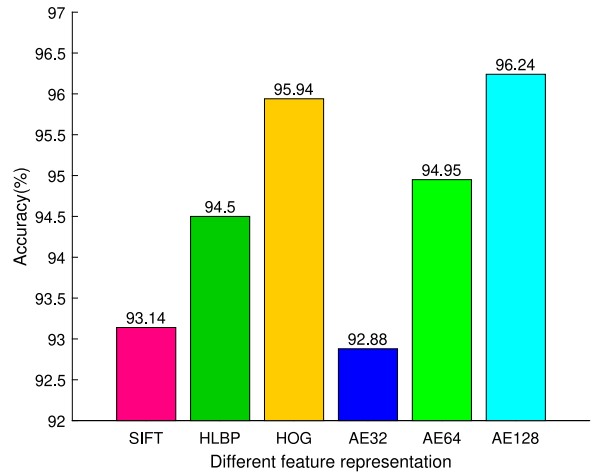


Fig. 9. Accuracies of different feature representations on RaFD dataset based on MiniTR.

4.8. NetScore analysis of different lightweight deep models

The NetScore is a better universal metric for evaluating deep neural network design. We compare some lightweight deep models, such as DDRGCN, MicroExpNet and three kinds of our proposed models. The results are shown in Fig. 10.

The higher the Netscore, the better the network design. From the results shown in Fig. 10, the proposed model Mini-baseline obtains the highest NetScore. But from the perspective of accuracy, Mini-baseline obtains 93.92% on RaFD dataset and is lower than other lightweight deep models, such as DDRGCN, Mini-one-head and MiniTR. Although the NetScore of the MiniTR is lower than Mini-baseline, the MiniTR obtains the best accuracy of 95.94%. Moreover, the score difference between Mini-baseline and MiniTR is relatively small. Compared to DDRGCN and MicroExpNet, our proposed MiniTR gets better performance in terms of NetScore and accuracy. With full consideration of overall performance, we adopt the MiniTR as the backbone of our proposed binary deep network.

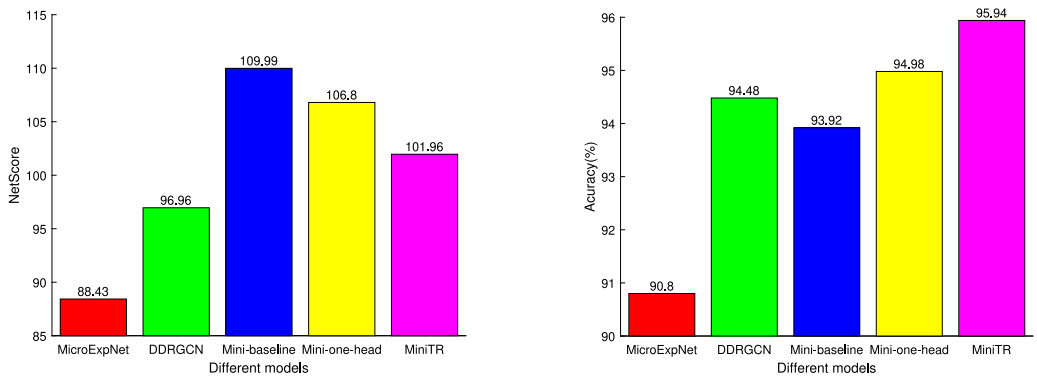
5. Conclusions and future work

In this paper, we design a lightweight binary transformer deep network for FER. The binary operation can improve the efficiency of the deep models for their deployment on resource-constrained platforms, and the transformer offers strong potential for capturing long-range interactions between facial muscle movements. To locate facial muscle movements, we adopt facial ROIs associated with AUs, and introduce the HOG feature to represent these ROIs which are used to construct patch-based tokens. Furthermore, we design a novel deep network named MiniTR based on Convs and transformer encoder. The parameters of the MiniTR are 69K and the transformer encoder can effectively mine the long-range muscle movements to improve the performance. All the experiments confirm that the proposed MiniTR obtains higher performance than other lightweight deep models. Finally, we propose a binary model MiniBTR by extending the MiniTR based on binarization, and the MiniBTR still achieves superior performance compared to other lightweight binary neural networks.

In future work, we will focus on the following two aspects: (1) With the popularity of various sensors, the acquisition of multimodal information is becoming more convenient. Utilizing the complementary advantages of multimodal information can further improve the performance of facial expression recognition. So designing an efficient

Table 8
Speed of different models on RaFD dataset.

Model	I7_4790 (ms)	GTX1070 (ms)
StudentExpNet_M (Cugu et al., 2019)	5.75	1.24
StudentExpNet_S (Cugu et al., 2019)	5.70	1.17
StudentExpNet_XS (Cugu et al., 2019)	5.65	1.14
L3Net (Cheng et al., 2020)	11.68	2.16
DDRGCN (Jin et al., 2021)	3.45	1.15
MicroExpNet (Cugu et al., 2019)	5.61	1.11
Mini-baseline	2.91	0.88
Mini-one-head	2.96	0.89
MiniTR	3.06	0.92



(a) NetScore of different models (b) Accuracies of different models

Fig. 10. NetScore and corresponding recognition accuracy of different models on RaFD dataset.

multi-modal fusion deep model is a good research direction. (2) Our proposed MiniTR obtain higher performance in the frontal facial expression analysis task. When there are complex scenes in the dataset, the proposed model obtains low accuracy. So designing a more robust FER framework for the wild environment is becoming a promising direction.

CRediT authorship contribution statement

Xing Jin: Writing – original draft, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Xiyin Wu:** Writing – review & editing, Visualization, Conceptualization. **Libo Weng:** Writing – review & editing, Validation, Resources, Investigation. **Qiaolin Ye:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by Guizhou Provincial Basic Research Program (Natural Science) (No. QianKeHe Basic - [2024] Youth 094).

Data availability

Data will be made available on request.

References

Altaf, M.F., Iqbal, M.W., Ali, G., Shinan, K., Alhazmi, H.E., Alanazi, F., Ashraf, M.U., 2025. Neural network-based ensemble approach for multi-view facial expression recognition. PloS One 20 (3), e0316562.

Bai, M., Yu, X., 2025. Lightweight emotion analysis solution using tiny machine learning for portable devices. Comput. Electr. Eng. 123, 110038.

Bicer, E., Kose, H., 2024. LITE-FER: A lightweight facial expression recognition framework for children in resource-limited devices. In: 2024 IEEE 18th International Conference on Automatic Face and Gesture Recognition. FG, IEEE, pp. 1–9.

Bougourzi, F., Dornaika, F., Mokrani, K., Taleb-Ahmed, A., Ruichek, Y., 2020. Fusing transformed deep and shallow features (FTDS) for image-based facial expression recognition. Expert Syst. Appl. 156, 113459.

Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J.D., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., et al., 2020. Language models are few-shot learners. Adv. Neural Inf. Process. Syst. 33, 1877–1901.

Chen, Y., Li, K., Tian, F., Wei, G., Seberi, M., 2025a. Lightweight expression recognition combined attention fusion network with hybrid knowledge distillation for occluded e-learner facial images. Neurocomputing 628, 129656.

Chen, Y., Peng, C., Wang, X., Zheng, Y., 2025b. Self-learning weight network based on label distribution training for facial expression recognition. IET Image Process. 19 (1), e13326.

Chen, X., Yan, B., Zhu, J., Wang, D., Yang, X., Lu, H., 2021. Transformer tracking. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 8126–8135.

Cheng, X., Miao, Z., Qiu, Q., 2020. Graph neural network with low-rank learnable local filters. arXiv:2008.01818 URL: https://arxiv.org/abs/2008.01818.

Cheng, B., Schwing, A., Kirillov, A., 2021. Per-pixel classification is not all you need for semantic segmentation. Adv. Neural Inf. Process. Syst. 34.

Cugu, I., Sener, E., Akbas, E., 2019. MicroExpNet: An extremely small and fast model for expression recognition from face images. In: 2019 Ninth International Conference on Image Processing Theory, Tools and Applications. IPTA, IEEE, pp. 1–6.

Danelakis, A., Theoharis, T., Pratikakis, I., 2018. Action unit detection in 3 d facial videos with application in facial expression retrieval and recognition. Multimedia Tools Appl. 77 (19), 24813–24841.

Ding, H., Zhou, S.K., Chellappa, R., 2017. Facenet2expnet: Regularizing a deep face recognition net for expression recognition. In: 2017 12th IEEE International Conference on Automatic Face & Gesture Recognition (FG 2017). IEEE, pp. 118–126.

Ekman, R., 1997. What the Face Reveals: Basic and Applied Studies of Spontaneous Expression Using The Facial Action Coding System (FACS). Oxford University Press, USA.

Fan, Y., Lam, J.C., Li, V.O., 2018. Multi-region ensemble convolutional neural network for facial expression recognition. In: International Conference on Artificial Neural Networks. Springer, pp. 84–94.

Ferro-Pérez, R., Mitre-Hernandez, H., 2020. ResMoNet: A residual mobile-based network for facial emotion recognition in resource-limited systems. arXiv preprint arXiv:2005.07649.

Ghimire, D., Jeong, S., Lee, J., Park, S.H., 2017. Facial expression recognition based on local region specific features and support vector machines. Multimedia Tools Appl. 76 (6), 7803–7821.

Grover, R., Bansal, S., 2025. Enhancing facial expression recognition in uncontrolled environment: a lightweight CNN approach with pre-processing. Neural Comput. Appl. 1–16.

Gursesli, M.C., Lombardi, S., Duradoni, M., Bocchi, L., Guazzini, A., Lanata, A., 2024. Facial emotion recognition (FER) through custom lightweight CNN model: performance evaluation in public datasets. IEEE Access.

Hazourli, A.R., Djeghri, A., Salam, H., Othmani, A., 2020. Deep multi-facial patches aggregation network for facial expression recognition. arXiv preprint arXiv:2002.09298.

Hubara, I., Courbariaux, M., Soudry, D., El-Yaniv, R., Bengio, Y., 2016. Binarized neural networks. In: Proceedings of the 30th International Conference on Neural Information Processing Systems. pp. 4114–4122.

Hussein, H.I., Dino, H.I., Mstafa, R.J., Hassan, M.M., 2022. Person-independent facial expression recognition based on the fusion of HOG descriptor and cuttlefish algorithm. Multimedia Tools Appl. 81 (8), 11563–11586.

Ioffe, S., Szegedy, C., 2015. Batch normalization: Accelerating deep network training by reducing internal covariate shift. arXiv:1502.03167 URL: https://arxiv.org/pdf/1502.03167v3.

Jain, D.K., Shamsolmoali, P., Sehdev, P., 2019. Extended deep neural network for facial emotion recognition. Pattern Recognit. Lett. 120, 69–74.

Jan, A., Ding, H., Meng, H., Chen, L., Li, H., 2018. Accurate facial parts localization and deep learning for 3d facial expression recognition. In: 2018 13th IEEE International Conference on Automatic Face & Gesture Recognition (FG 2018). IEEE, pp. 466–472.

Jiang, B., Li, N., Cui, X., Liu, W., Yu, Z., Xie, Y., 2024. Research on facial expression recognition algorithm based on lightweight transformer. Inform. 15 (6), 321.

Jin, X., Lai, Z., Jin, Z., 2021. Learning dynamic relationships for facial expression recognition based on graph convolutional network. IEEE Trans. Image Process. 30, 7143–7155.

Jin, X., Sun, W., Jin, Z., 2020. A discriminative deep association learning for facial expression recognition. Int. J. Mach. Learn. Cybern. 11 (4), 779–793.

Jingying, C., Jinxin, S., Ruyi, X., 2024. Dual subspace manifold learning based on GCN for intensity-invariant facial expression recognition. Pattern Recognit. 148.

Jung, H., Lee, S., Yim, J., Park, S., Kim, J., 2015. Joint fine-tuning in deep neural networks for facial expression recognition. In: Proceedings of the IEEE International Conference on Computer Vision. pp. 2983–2991.

Khorrami, P., Paine, T., Huang, T., 2015. Do deep neural networks learn facial action units when doing expression recognition? In: Proceedings of the IEEE International Conference on Computer Vision Workshops. pp. 19–27.

Kumar, S., Bhuyan, M.K., Chakraborty, B.K., 2016. Extraction of informative regions of a face for facial expression recognition. IET Comput. Vis. 10 (6), 567–576.

Kuo, C., Lai, S., Sarkis, M., 2018. A compact deep learning model for robust facial expression recognition. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. pp. 2121–2129.

Kuo, C., Lai, S., Sarkis, M., 2018b. A compact deep learning model for robust facial expression recognition. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. pp. 2121–2129.

Lan, X., Xue, J., Qi, J., Jiang, D., Lu, K., Chua, T.-S., 2025. Expllm: Towards chain of thought for facial expression recognition. IEEE Trans. Multimed..

Langner, O., Dotsch, R., Bijlstra, G., Wigboldus, D.H., Hawk, S.T., Van Knippenberg, A., 2010. Presentation and validation of the radboud faces database. Cogn. Emot. 24 (8), 1377–1388.

Lee, J.R.H., Wang, L., Wong, A., 2020. EmotionNet nano: An efficient deep convolutional neural network design for real-time facial expression recognition. arXiv preprint arXiv:2006.15759.

Li, S., Deng, W., 2018. Reliable crowdsourcing and deep locality-preserving learning for unconstrained facial expression recognition. IEEE Trans. Image Process. 28 (1), 356–370.

Li, S., Deng, W., Du, J., 2017. Reliable crowdsourcing and deep locality-preserving learning for expression recognition in the wild. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. pp. 2852–2861.

Li, Y., Lu, Y., Chen, B., Zhang, Z., Li, J., Lu, G., Zhang, D., 2021. Learning informative and discriminative features for facial expression recognition in the wild. IEEE Trans. Circuits Syst. Video Technol. 32 (5), 3178–3189.

Liao, L., Wu, S., Song, C., Fu, J., 2024. RS-xception: A lightweight network for facial expression recognition. Electronics 13 (16), 3217.

Liu, C., Chen, L.-C., Schroff, F., Adam, H., Hua, W., Yuille, A.L., Fei-Fei, L., 2019. Auto-deeplab: Hierarchical neural architecture search for semantic image segmentation. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 82–92.

Liu, T., Li, J., Wu, J., Du, B., Chang, J., Liu, Y., 2023. Facial expression recognition on the high aggregation subgraphs. IEEE Trans. Image Process. 32, 3732–3745.

Liu, Z., Lin, Y., Cao, Y., Hu, H., Wei, Y., Zhang, Z., Lin, S., Guo, B., 2021. Swin transformer: Hierarchical vision transformer using shifted windows. In: Proceedings of the IEEE/CVF International Conference on Computer Vision. pp. 10012–10022.

Liu, M., Shan, S., Wang, R., Chen, X., 2014. Learning expressionlets on spatio-temporal manifold for dynamic facial expression recognition. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. pp. 1749–1756.

Liu, Z., Wu, B., Luo, W., Yang, X., Liu, W., Cheng, K.-T., 2018. Bi-real net: Enhancing the performance of 1-bit cnns with improved representational capability and advanced training algorithm. In: Proceedings of the European Conference on Computer Vision. ECCV, pp. 722–737.

Lu, L., Jiang, Y., Li, X., Wang, H., Zou, Q., Wang, Q., 2025. Depression and anxiety detection method based on serialized facial expression imitation. Eng. Appl. Artif. Intell. 149, 110354.

Lucey, P., Cohn, J.F., Kanade, T., Saragih, J., Ambadar, Z., Matthews, I., 2010. The extended cohn-kanade dataset(ck+): A complete dataset for action unit and emotion-specified expression. In: IEEE Computer Society Conference on Computer Vision Ans Pattern Recognition Workshops. pp. 94–101.

Mao, S., Li, X., Zhang, F., Peng, X., Yang, Y., 2025. Facial action units as a joint dataset training bridge for facial expression recognition. IEEE Trans. Multimed..

Misra, I., Girdhar, R., Joulin, A., 2021. An end-to-end transformer model for 3D object detection. In: Proceedings of the IEEE/CVF International Conference on Computer Vision. pp. 2906–2917.

Mollahosseini, A., Chan, D., Mahoor, M.H., 2016. Going deeper in facial expression recognition using deep neural networks. In: 2016 IEEE Winter Conference on Applications of Computer Vision. WACV, IEEE, pp. 1–10.

Pan, B., Hirota, K., Dai, Y., Jia, Z., Shao, S., She, J., 2025. Learning sequential variation information for dynamic facial expression recognition. IEEE Trans. Neural Networks Learn. Syst..

Qin, H., Gong, R., Liu, X., Bai, X., Song, J., Sebe, N., 2020. Binary neural networks: A survey. Pattern Recognit. 105, 107281.

Raghu, M., Unterthiner, T., Kornblith, S., Zhang, C., Dosovitskiy, A., 2021. Do vision transformers see like convolutional neural networks? Adv. Neural Inf. Process. Syst. 34.

Rastegari, M., Ordonez, V., Redmon, J., Farhadi, A., 2016. Xnor-net: Imagenet classification using binary convolutional neural networks. In: European Conference on Computer Vision. Springer, pp. 525–542.

Shan, C., Gong, S., McOwan, P.W., 2005. Recognizing facial expressions at low resolution. In: IEEE Conference on Advanced Video and Signal Based Surveillance, 2005.. IEEE, pp. 330–335.

Srinivas, A., Lin, T.-Y., Parmar, N., Shlens, J., Abbeel, P., Vaswani, A., 2021. Bottleneck transformers for visual recognition. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 16519–16529.

Sun, X., Xia, P., Ren, F., 2021. Multi-attention based deep neural network with hybrid features for dynamic sequential facial expression recognition. Neurocomputing 444, 378–389.

Sun, W., Zhao, H., Jin, Z., 2017. An efficient unconstrained facial expression recognition algorithm based on stack binarized auto-encoders and binarized neural networks. Neurocomputing 267, 385–395.

Sun, W., Zhao, H., Jin, Z., 2018. A visual attention based ROI detection method for facial expression recognition. Neurocomputing 296, 12–22.

Sutskever, I., Vinyals, O., Le, Q.V., 2014. Sequence to sequence learning with neural networks. Adv. Neural Inf. Process. Syst. 27.

Tang, Y., Zhang, X., Hu, X., Wang, S., Wang, H., 2020. Facial expression recognition using frequency neural network. IEEE Trans. Images Process. 30, 444–457.

Touvron, H., Cord, M., Douze, M., Massa, F., Sablayrolles, A., Jégou, H., 2021. Training data-efficient image transformers & distillation through attention. In: International Conference on Machine Learning. PMLR, pp. 10347–10357.

Tutuianu, G.I., Liu, Y., Alamäki, A., Kauttonen, J., 2024. Benchmarking deep facial expression recognition: An extensive protocol with balanced dataset in the wild. Eng. Appl. Artif. Intell. 136, 108983.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł., Polosukhin, I., 2017. Attention is all you need. Adv. Neural Inf. Process. Syst. 30.

Wang, J., Li, X., Ling, C.X., 2018. Pelee: A real-time object detection system on mobile devices. In: Advances in Neural Information Processing Systems. pp. 1967–1976.

Wang, W., Xie, E., Li, X., Fan, D.-P., Song, K., Liang, D., Lu, T., Luo, P., Shao, L., 2021. Pyramid vision transformer: A versatile backbone for dense prediction without convolutions. In: Proceedings of the IEEE/CVF International Conference on Computer Vision. pp. 568–578.

Wang, J., Zhang, Z., 2023. Facial expression recognition in online course using lightweight vision transformer via knowledge distillation. In: Pacific Rim International Conference on Artificial Intelligence. Springer, pp. 247–253.

Wong, A., 2019. NetScore: towards universal metrics for large-scale performance analysis of deep neural networks for practical on-device edge usage. In: International Conference on Image Analysis and Recognition. Springer, pp. 15–26.

Wong, A., Shafiee, M.J., Chwyl, B., Li, F., 2018. Ferminets: Learning generative machines to generate efficient neural networks via generative synthesis. arXiv preprint arXiv:1809.05989.

Wu, H., Xiao, B., Codella, N., Liu, M., Dai, X., Yuan, L., Zhang, L., 2021. Cvt: Introducing convolutions to vision transformers. In: Proceedings of the IEEE/CVF International Conference on Computer Vision. pp. 22–31.

Xie, Y., Tian, W., Yu, Z., 2023. Robust facial expression recognition with transformer block enhancement module. Eng. Appl. Artif. Intell. 126, 106795.

Xiong, Y.-J., Wang, Q., Du, Y., Lu, Y., 2024. Adaptive graph-based feature normalization for facial expression recognition. Eng. Appl. Artif. Intell. 129, 107623.

Ye, Q., Yang, J., Liu, F., Zhao, C., Ye, N., Yin, T., 2016. L1-norm distance linear discriminant analysis based on an effective iterative algorithm. IEEE Trans. Circuits Syst. Video Technol. 28 (1), 114–129.

Zeng, M., Luo, Y., Liu, G., 2023. Lightweight facial expression recognition network with dynamic deep mutual learning. In: Proceedings of the 2023 3rd International Conference on Bioinformatics and Intelligent Computing. pp. 222–226.

Zhang, K., Li, Y., Wang, J., Cambria, E., Li, X., 2021. Real-time video emotion recognition based on reinforcement learning and domain knowledge. IEEE Trans. Circuits Syst. Video Technol. 32 (3), 1034–1047.

Zhang, X., Lu, Y., Yan, H., Huang, J., Gu, Y., Ji, Y., Liu, Z., Liu, B., 2025. Resup: Reliable label noise suppression for facial expression recognition. IEEE Trans. Affect. Comput..

Zhao, S., Cai, H., Liu, H., Zhang, J., Chen, S., 2018. Feature selection mechanism in CNNs for facial expression recognition.. In: British Machine Vision Conference. pp. 1–12.

Zhao, G., Huang, X., Taini, M., Li, S.Z., PietikäInen, M., 2011. Facial expression recognition from near-infrared videos. Image Vis. Comput. 29 (9), 607–619.

Zhao, X., Liang, X., Liu, L., Li, T., Han, Y., Vasconcelos, N., Yan, S., 2016. Peak-piloted deep network for facial expression recognition. In: European Conference on Computer Vision. Springer, pp. 425–442.

Zhou, Y., Pei, J., Si, W., Qin, J., Heng, P.-A., 2025. Delving into quaternion wavelet transformer for facial expression recognition in the wild. IEEE Trans. Multimed..

Zhou, J., Zhang, X., Liu, Y., Lan, X., 2020. Facial expression recognition using spatial-temporal semantic graph network. In: 2020 IEEE International Conference on Image Processing. ICIP, IEEE, pp. 1961–1965.

Zhu, Y., Ai, J., Xue, W., Wu, M., Yang, S., Jia, W., Hu, M., 2025. Cross-domain facial expression recognition: Bi-directional fusion of active and stable information. Eng. Appl. Artif. Intell. 149, 110357.

Zhu, S., Dong, X., Su, H., 2019. Binary ensemble neural network: More bits per network or more networks per bit? In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 4923–4932.

Zhu, F., Gao, J., Yang, J., Ye, N., 2022. Neighborhood linear discriminant analysis. Pattern Recognit. 123, 108422.